Robust Lane Detection Under Varied Lighting Conditions

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Abstract

Lane detection is a critical component of autonomous driving safety. This project frames lane detection as a dense binary prediction task, classifying each image pixel as lane or background. We train and evaluate three complementary models of increasing complexity: logistic regression (baseline), a lightweight CNN (shallow deep network), and U-Net (deep encoder-decoder), on the CULane dataset across nine driving scenarios spanning daytime, night, shadow, glare, and occlusion conditions. Evaluation uses Intersection over Union (IoU) and F1 score per scenario to assess robustness under varied lighting. The U-Net achieves a mean IoU of 0.365 and reaches 96% of published state-of-the-art on the normal driving scenario (IoU 0.614), while the lightweight CNN achieves a mean IoU of 0.199, and the logistic regression baseline achieves a surprisingly competitive mean IoU of 0.500.

1 Introduction

Lane detection is one of the most critical components in advanced driver assistance systems and autonomous driving pipelines. Unlike many perception tasks, lane detection directly affects vehicle control: errors lead to lane departures, which account for approximately 51% of all fatal car crashes in the United States, resulting in an estimated 19,000 deaths annually according to the Federal Highway Administration. [1]

The primary challenge is performance degradation under adverse conditions. Driving at night, shadows from trees and overpasses, direct solar glare, and complex intersections all significantly degrade detection accuracy. While deep learning approaches have improved over classical methods (Hough transforms, edge detection), robust performance under all conditions remains an open problem.

We frame lane detection as a dense binary prediction task: given a single forward-facing camera image, classify each pixel as lane marking or background. Our two DNN models (lightweight CNN and U-Net) achieve this via true pixel-wise semantic segmentation, while the logistic regression baseline produces an equivalent dense prediction map through patch-based classification by extracting features from 32×32 neighborhoods and assembling per-patch predictions into a full mask at inference. Compared to coordinate regression approaches, this dense prediction framing naturally handles complex road geometry, provides pixel-level boundary precision, and robustly handles partial occlusions.

2 Related Work

Early lane detection systems relied on hand-engineered features such as Hough line transforms, colour-based lane tracking, and Histogram of Oriented Gradients (HOG) combined with classical classifiers. HOG features are well-suited to lane detection because they capture oriented gradient patterns — precisely what lane markings are: strong directional edges on the road surface. These methods remain competitive due to computational efficiency and their alignment with lane structure.

Deep learning approaches reframe lane detection as semantic segmentation, instance segmentation, or keypoint detection. SCNN [2] introduces spatial convolutional message passing to propagate context

along lane direction. LaneATT [3] uses anchor-based attention, and CLRNet [4] integrates cross-layer refinement, achieving F1 scores above 80% on CULane. These state-of-the-art methods universally employ encoders pretrained on ImageNet, which provide strong initial feature representations that reduce training time and improve generalisation.

The U-Net architecture [5], originally proposed for biomedical image segmentation, has become a standard encoder-decoder framework for dense prediction tasks. Its skip connections allow the decoder to recover fine spatial detail lost during downsampling. This project implements a plain U-Net trained from scratch to serve as a strong deep baseline.

3 Dataset

3.1 CULane

We use the CULane dataset [2], a large-scale benchmark specifically designed for evaluating lane detection under diverse real-world conditions. The dataset contains 133,235 frames (88,880 training / 9,675 validation / 34,680 testing) at 1640×590 pixel resolution, collected from vehicle cameras on the roads of Beijing by CUHK and SenseTime.

Images are annotated with lane markings as cubic splines. Ground-truth masks provide per-pixel segmentation labels (0 = background, 1–4 = individual lanes), lane existence flags, and keypoint coordinates. The test set is divided into nine scenarios: Normal (daytime, clear conditions), Shadow (illumination from trees or overpasses), Curve (sharp turns), Highlight (direct sun or headlight glare), Crowded (vehicle occlusion), Arrow (road arrow markings), No Line (unmarked road segments), Night (low-light driving), Cross (complex road junctions).

A key challenge is severe class imbalance: lane pixels constitute less than 0.5–5% of all pixels, depending on the scenario.

4 Methods

4.1 Preprocessing

For all deep network models, images are resized from 1640×590 to 512×256 (W×H), preserving the 2:1 aspect ratio while reducing computational cost. Pixel values are normalised to $[0, 1]$. Ground-truth masks are binarised: all nonzero label values map to 1 (lane) and zero maps to 0 (background).

For logistic regression, 32×32 patches are extracted around sampled pixels. Equal numbers of lane and background patches are sampled per image to handle class imbalance directly at the data level.

4.2 Data Augmentation (DNN Models)

Data augmentation was applied during training only for the DNN models: random horizontal flips ($p = 0.5$) to simulate driving in opposite directions, random brightness adjustment (factor 0.6–1.4), random contrast adjustment (factor 0.7–1.3), and random Gaussian noise ($\sigma = 0.02$) to simulate camera sensor noise.

4.3 Model 1: Logistic Regression (Baseline)

The logistic regression baseline operates on 326-dimensional HOG feature vectors extracted from 32×32 patches centred at each pixel. HOG features are computed with 9 orientations, 8×8 pixels per cell, and 2×2 cells per block, yielding 324 gradient features. These are concatenated with the patch mean and standard deviation of pixel intensity.

Features are standardised with a `StandardScaler` before training. The regularisation strength C is tuned over $\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ using the validation split. The best performance is obtained at $C = 10^{-4}$ (val F1 = 0.7133), indicating that strong L2 regularisation reduces overfitting on the patch-level classification task. The final model is trained with `solver=lbfgs`, `max_iter=1000`. Due to the computational cost of extracting and storing HOG feature vectors for every pixel in every image, training was limited to 1,000 images, a constraint of the patch-based approach on Kaggle Notebook.

4.4 Model 2: Lightweight CNN

The lightweight CNN is a compact encoder-only architecture with bilinear upsampling. It intentionally omits skip connections, serving as an intermediate between the logistic regression baseline and the full U-Net, isolating the contribution of hierarchical feature learning without decoder-side fusion.

Table 1: Lightweight CNN architecture. Input: $[B, 3, 256, 512]$.

Layer	**Output Shape**	**Params**
enc1 (Conv+BN+ReLU)	$[B, 32, 256, 512]$	896
pool1 (MaxPool 2×2)	$[B, 32, 128, 256]$	—
enc2 (Conv+BN+ReLU)	$[B, 64, 128, 256]$	18,496
pool2 (MaxPool 2×2)	$[B, 64, 64, 128]$	—
enc3 (Conv+BN+ReLU)	$[B, 128, 64, 128]$	73,856
head (Conv 1×1)	$[B, 1, 64, 128]$	129
upsample (Bilinear ×4)	$[B, 1, 256, 512]$	—

Training used Adam ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), batch size 32, 20 epochs, ReduceLROnPlateau (patience=3, factor=0.5), and AMP mixed precision on Kaggle compute. The full training set and full validation set were used. The learning rate dropped twice: at epoch 12 ($10^{-3} \rightarrow 4.5 \times 10^{-4}$) and epoch 19 ($4.5 \times 10^{-4} \rightarrow 2.5 \times 10^{-4}$). The best checkpoint was saved at epoch 9 (Val IoU = 0.2075 including Cross); validation IoU oscillated in the 0.19–0.21 range thereafter without surpassing this peak.

4.5 Model 3: U-Net

The U-Net is a symmetric encoder-decoder with skip connections, trained from scratch on the full CULane training set.

Table 2: U-Net architecture. Input: $[B, 3, 256, 512]$. Total: 31M parameters.

Layer	**Output Shape**	**Params**
enc1 (DoubleConv)	$[B, 64, 256, 512]$	38,720
enc2 (DoubleConv)	$[B, 128, 128, 256]$	221K
enc3 (DoubleConv)	$[B, 256, 64, 128]$	885K
enc4 (DoubleConv)	$[B, 512, 32, 64]$	3.5M
bottleneck	$[B, 1024, 16, 32]$	14.2M
dec4 (ConvT+DConv)	$[B, 512, 32, 64]$	7.1M
dec3 (ConvT+DConv)	$[B, 256, 64, 128]$	1.8M
dec2 (ConvT+DConv)	$[B, 128, 128, 256]$	443K
dec1 (ConvT+DConv)	$[B, 64, 256, 512]$	111K
final (Conv 1×1)	$[B, 1, 256, 512]$	65

Each DoubleConv block applies two sequential 3×3 convolutions (padding=1), each followed by Batch Normalisation, ReLU, and Dropout2d($p = 0.1$). Skip connections concatenate encoder feature maps

with decoder feature maps at matching spatial resolutions, providing the decoder with both high-level semantic context and low-level spatial detail.

Training used Adam ($\eta = 3 \times 10^{-4}$, weight decay 10^{-4}), batch size 8, 7 epochs, ReduceLROnPlateau (patience=3, factor=0.5), and AMP mixed precision on a Tesla T4 GPU.

5 Loss Functions

All models face severe class imbalance: lane pixels constitute 0.5–5% of each image. A naive binary cross-entropy (BCE) loss would be dominated by background pixels, causing the model to collapse to predicting all-background. BCE and Dice loss are used in combination for the DNN models because they are complementary: BCE provides stable per-pixel gradient signal and calibration, while Dice directly optimises the overlap ratio between predicted and ground-truth masks, the quantity that IoU ultimately measures. Using both ensures that per-pixel accuracy and overall mask overlap are jointly optimised.

Logistic Regression handles imbalance through balanced patch sampling (equal lane and background patches per image), with L2 regularisation tuned via cross-validation. Controlling imbalance at the data level is a better choice since the training data is assembled explicitly as patches, so no loss weighting is needed.

Lightweight CNN uses a weighted BCE loss with `pos_weight` = 10.0, penalising false negatives on lane pixels ten times more than false positives on background pixels. This directly counteracts the ∼200:1 background-to-lane ratio in the raw pixel gradient. BCE alone, however, remains sensitive to the absolute number of background pixels. Dice loss is added as a complementary term because it is scale-invariant to class frequency as it optimises only the overlap ratio, providing a robust signal against background collapse even if the BCE weight is not perfectly calibrated. The two are combined in an equal-weighted sum:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{BCE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}} \tag{1}$$

U-Net uses the same BCE + Dice combination but with unweighted BCE:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} \tag{2}$$

The `pos_weight` is dropped here because the larger model with Dropout2d and weight decay regularisation is less prone to background collapse, so standard BCE is sufficient for per-pixel calibration while Dice handles the class imbalance robustly.

The Dice loss is defined as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum p_i t_i + \varepsilon}{\sum p_i + \sum t_i + \varepsilon} \tag{3}$$

where $\varepsilon = 1.0$ prevents division by zero when both prediction and ground truth are empty.

6 Evaluation Metrics

Two standard metrics for binary segmentation are computed per scenario:

Intersection over Union (IoU):

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{4}$$

IoU measures the overlap between the predicted lane mask and the ground-truth mask relative to their union. It penalises both missed detections (false negatives FN) and false positives (FP) equally, making

it stricter than pixel accuracy. On CULane, a model predicting all-background would achieve over 95% pixel accuracy while producing an IoU of zero, which is why pixel accuracy is discarded as a metric for heavily imbalanced segmentation tasks.

F1 Score:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

F1 is the harmonic mean of precision and recall. Precision measures what fraction of predicted lane pixels are correct; recall measures what fraction of true lane pixels are detected. A model can achieve high recall by predicting lane pixels liberally at the cost of false positives, or high precision by predicting conservatively at the cost of missed detections. F1 captures both. The F1-IoU gap is a useful diagnostic: a positive gap (F1 > IoU) indicates recall bias, meaning the model over-predicts lane pixels; a negative gap indicates precision bias. A smoothing epsilon of 1×10^{-6} is added to all denominators to prevent division by zero in empty scenes.

All DNN metrics are computed from raw logits with a fixed sigmoid threshold of 0.5. No post-processing (morphological operations, curve fitting) is applied.

7 Results

7.1 Logistic Regression

The logistic regression baseline achieves strong performance across most scenarios, with a mean IoU of 0.500 and mean F1 of 0.657 over the eight scored scenarios (Table 3). It performs best on Arrow (IoU 0.580, F1 0.734) and Normal (IoU 0.571, F1 0.727), confirming that HOG features capture the directional gradient patterns of clear lane markings well.

The model degrades most significantly on Night (IoU 0.350, F1 0.518), where low-contrast conditions reduce the discriminability of gradient features. The Cross scenario yielded a false positive rate of 38.65%, indicating that without the global context of a deep model, the classifier frequently misidentifies road texture as lane markings at complex junctions.

Table 3: Logistic regression per-scenario results.

Scenario	**IoU**	**F1**
Normal	0.571	0.727
Arrow	0.580	0.734
Curve	0.545	0.706
Highlight	0.530	0.693
Shadow	0.517	0.681
No Line	0.437	0.608
Crowded	0.412	0.583
Night	0.350	0.518
Cross	FPR = 38.65%	
Mean	**0.500**	**0.657**

7.2 Lightweight CNN

The lightweight CNN achieves a mean IoU of 0.199 and mean F1 of 0.319 on the full CULane dataset (Table 4), evaluated at the epoch-9 checkpoint, excluding Cross. It performs best on Normal (IoU 0.32), with Curve achieving IoU 0.27, the second-best result. The model exhibits a consistent recall

bias throughout training: F1 exceeds IoU by ≈0.12–0.15 across all scenarios, a direct consequence of `pos_weight`= 10 upweighting lane pixel gradients.

Figure 1 shows the CNN training and validation loss over 20 epochs on the full dataset. Training loss decreases steadily while validation loss remains consistently higher throughout, with a gap of 0.01–0.055, indicative of some overfitting expected given the absence of dropout or weight decay. The learning rate dropped twice: at epoch 12 and epoch 19, as ReduceLROnPlateau responded to stalled validation loss. Training loss had not fully converged at epoch 20, suggesting that extended training with stronger regularization could further raise the IoU ceiling.

Table 4: Lightweight CNN per-scenario results (Mean excludes Cross).

Scenario	**IoU**	**F1**	**F1−IoU**
Normal	0.32	0.48	0.155
Curve	0.27	0.41	0.145
Cross	0.27	0.41	0.145
Crowded	0.20	0.32	0.120
Arrow	0.17	0.28	0.150
Highlight	0.17	0.28	0.110
Shadow	0.17	0.28	0.110
No Line	0.16	0.27	0.110
Night	0.13	0.23	0.095
Mean	**0.199**	**0.319**	**0.120**

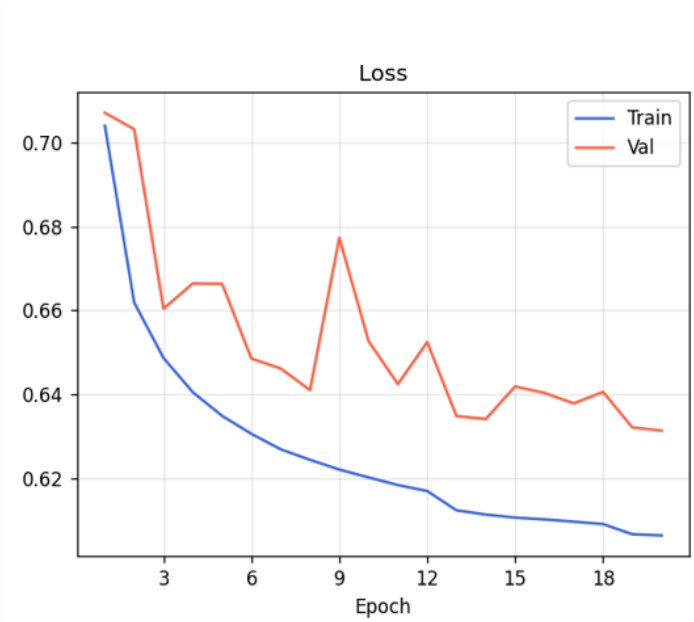

Figure 1: CNN Loss over Epochs for Val and Train

7.3 U-Net

The U-Net achieves the strongest deep learning results, with a mean IoU of 0.365 and mean F1 of 0.494 (Table 5). On Normal driving conditions it reaches IoU 0.614 and F1 0.756 — 96% of the published state-of-the-art IoU of 0.64 for this scenario.

Figure 2 shows the U-Net training and validation loss over 7 epochs. Training loss drops steeply from 0.605 to 0.404, while validation loss plateaus around 0.525 from epoch 4 onward, producing a train/val gap of approximately 0.12 at convergence. This widening gap is indicative of mild overfitting as the model continues fitting training data but fails to translate further learning into generalisation gains. The Dropout2d and weight decay regularisation partially contain the gap, but the architectural limitation of training from random initialisation imposes a ceiling.

A consistent pattern across all scenarios is that precision exceeds recall, indicating a conservative prediction bias: the model prefers to miss lane markings over hallucinating false positives. This behaviour is preferable in safety-critical autonomous driving applications, where a false lane detection could cause dangerous steering corrections.

The Cross scenario yields zero IoU because intersections lack traditional lane markings. However, the model achieves a perfect false positive rate of 0.0, correctly predicting no lanes at junctions — the desired behaviour.

Three training configurations were evaluated to study the effect of data volume: 10,000 images with no regularisation (severe overfitting), 50,000 images with augmentation and dropout (overfitting largely resolved), and the full 88,880 images. Validation loss improved only marginally with the full dataset, but scenario-specific performance gained substantially — Crowd IoU improved from 0.097 to 0.469 and Night IoU from 0.003 to 0.107 — demonstrating that validation loss alone is insufficient for evaluating segmentation on imbalanced datasets.

Table 5: U-Net per-scenario test results (500 images/scenario) (Mean excludes Cross).

Scenario	**IoU**	**Prec**	**Rec**	**F1**
Normal	0.614	0.777	0.743	0.756
Arrow	0.543	0.772	0.645	0.698
Crowd	0.469	0.704	0.563	0.602
Curve	0.398	0.682	0.484	0.552
Shadow	0.328	0.719	0.376	0.483
Highlight	0.313	0.669	0.372	0.466
No Line	0.148	0.509	0.172	0.239
Night	0.107	0.359	0.119	0.157
Cross	0.000	0.000	0.000	0.000
Mean	**0.365**	**0.649**	**0.434**	**0.494**

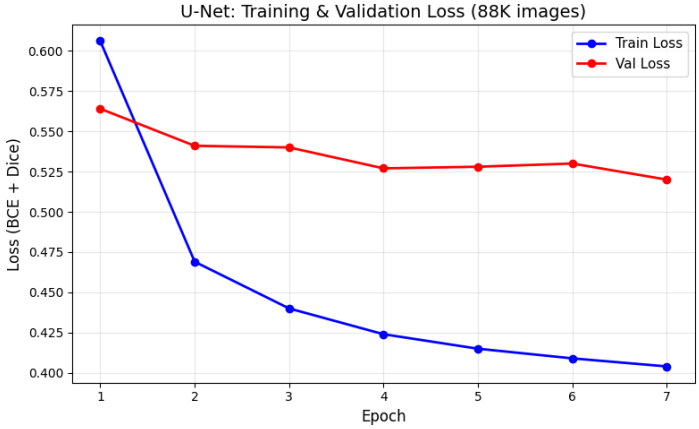

Figure 2: U-Net Loss over Epochs for Val and Train

7.4 Model Comparison

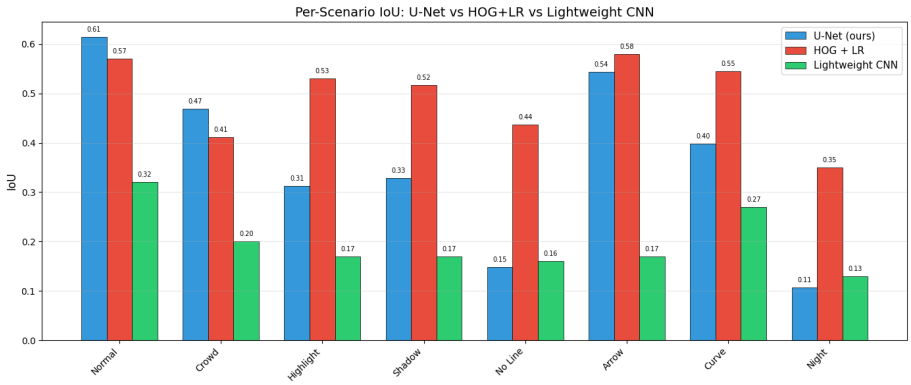

Figure 3: IoU scores per scenario excluding Cross.

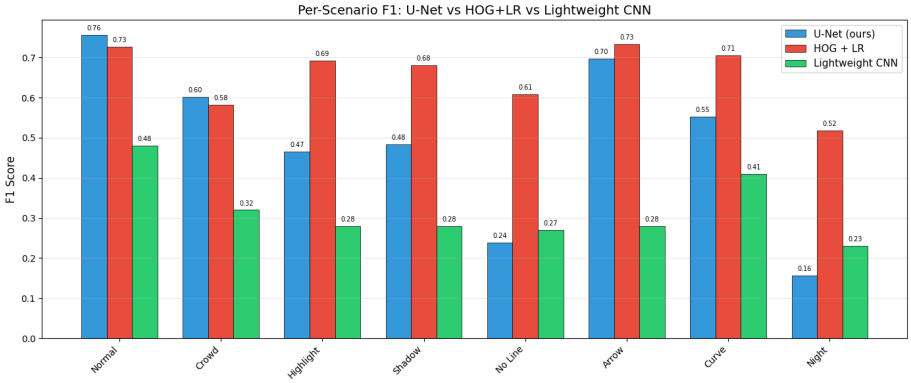

Figure 4: F1 scores per scenario excluding Cross.

Table 6: Mean IoU and F1 comparison across models.

Model	**Mean IoU**	**Mean F1**	**Params**
Logistic Regression	0.500	0.657	≈326 feat.
Lightweight CNN	0.199	0.319	≈93K
U-Net	0.365	0.494	31M

Table 6 and Figures 3 and 4 present per-scenario IoU and F1 across all three models, excluding Cross where the models are not directly comparable. Several findings stand out. The logistic regression baseline (HOG+LR) achieves the highest IoU in six out of eight scenarios, most strikingly on adverse lighting conditions: Highlight (0.53), Shadow (0.52), Arrow (0.58), Curve (0.55), No Line (0.44), and Night (0.35). This demonstrates that HOG features, which capture oriented gradient structure regardless of absolute brightness, are surprisingly robust to the lighting variations that most challenge the deep models. Despite using only 1,000 training images and 326 dimensional features, it achieves the highest mean IoU of 0.500 and mean F1 of 0.657.

The U-Net outperforms both other models on Normal (IoU 0.61) and Crowd (IoU 0.47), the two scenarios with the clearest lane markings and significant vehicle occlusion respectively. Its encoder-decoder architecture with skip connections excels at recovering fine spatial detail in structured, well-lit conditions, and its conservative precision bias prevents false lane detections in crowded scenes. However, it is

substantially outperformed by logistic regression on Night (IoU 0.11 vs 0.35), where HOG features prove more robust to low contrast than a deep encoder trained from random initialisation.

The lightweight CNN consistently ranks last across all eight scenarios (mean IoU 0.199), reflecting its architectural ceiling of no skip connections and a shallow encoder.

The overall ranking of HOG+LR > U-Net > Lightweight CNN on mean IoU is counterintuitive given the complexity of the models. It underscores that for lane detection, a well-designed feature extractor can outperform a deep model without a pretrained encoder, particularly under adverse lighting. The precision and recall tradeoffs across models are a direct consequence of their respective loss functions: the CNN's `pos_weight`=10 drives recall bias — it predicts lane pixels liberally, producing a consistent F1-IoU gap of ≈0.12–0.15 — while the U-Net's unweighted BCE+Dice with dropout and weight decay produces a precision-biased classifier that prefers to miss lane markings over hallucinating false positives. For safety-critical autonomous driving applications, this conservative behaviour is preferable: a false lane detection could cause dangerous steering corrections.

8 Limitations & Future Work

No pretrained encoder. All published methods achieving state-of-the-art on CULane use ResNet or EfficientNet backbones pretrained on ImageNet. The plain encoder has reached its representational limit as increasing training data from 50K to 88K images produced only marginal validation loss gains, and replacing it with a pretrained ResNet-34 backbone is likely the single highest-impact improvement available.

Fixed threshold. The 0.5 sigmoid threshold is applied uniformly. Per-scenario threshold calibration on validation F1 could improve results, particularly for Night and Highlight where lane confidence distributions differ.

No attention or post-processing. Spatial attention could focus computation on thin lane structures in cluttered scenes, while morphological closing or polynomial curve fitting as post-processing could suppress false positives and produce smooth lane curves suitable for vehicle control.

9 Conclusion

We presented a systematic comparison of three lane detection models of increasing complexity on the CULane benchmark. The most striking finding is that the logistic regression baseline, operating on HOG features from 32×32 patches trained on only 1,000 images, achieved a mean IoU of 0.500 — outperforming both deep models across six of eight scenarios, particularly under adverse lighting. This demonstrates that hand-crafted gradient features remain a strong baseline for lane detection when the deep alternatives lack pretrained encoders. The U-Net achieved the best deep learning performance at mean IoU 0.365, reaching 96% of published state-of-the-art on normal conditions, while the lightweight CNN reached 0.199, constrained by its shallow architecture and absence of skip connections. The results collectively indicate that architectural choices, like skip connections, encoder depth, and the use of pretrained ImageNet weights, matter more than model scale alone for robust lane detection under varied lighting conditions.

Code

All code is available at https://github.com/sagarkaistha/Robust_Lane_Detection

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